

ELSA DUBIL

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EDUCATION:

UCLA | 2018 - 2023

- Bachelor of Science in Bioengineering
- Awards: Engineering Achievement Award in Student Welfare, Will Rogers Scholarship

RELEVANT EXPERIENCE:

Illumina

Engineer 1 & 2 | November 2023 - Present

- Led software quality strategy across multiple cloud-based bioinformatics platforms used in large-scale genomic analysis, ensuring reliability, performance, and compliance across distributed systems.
- Developed a machine learning model that analyzed 800+ JIRA tickets to identify systemic component failures, reducing recurring bugs and optimizing lifecycle workflows.
- Authored clear, testable software requirements, validated test plans, and supervised verification activities to ensure traceability and uphold international standards.
- Managed Git-based version control, conducted code reviews, and drafted customer release notes and technical documentation for seamless deployment, version tracking, and accurate technical records.
- Led root cause investigations and partnered with DevOps, product, and cybersecurity teams to mitigate software risks and drive continuous product improvement.

Wong Laboratory, Undergraduate Research

Undergraduate Project Manager and Programming Lead | September 2021 - June 2023

- Spearheaded team of six in leveraging machine learning and variational autoencoding techniques to automate microscopic image analysis of over 50,000 images using C++, Python, and Julia programming languages, resulting in a 240% increase in image processing efficiency and optimizing data-driven insights.
- Led debugging efforts, developed and published comprehensive release notes and documentation, and led root cause analysis of bugs while implementing effective corrective action plans.

Lactation Lab Startup

Bioengineering Intern | August 2022 - September 2022

- Collaborated with a cross-functional team of software developers and scientists to ideate, design, and launch the world's first spectrophotometer smartphone application, which received FDA breakthrough device designation.

SKILLS:

- **Programming Languages:** Python, C++, JavaScript, TypeScript, HTML (advanced); MATLAB, Julia (intermediate)
- **Events and Projects:** Job Application Autofiller Chrome & Firefox Extension (2023), TreeHacks (Mental health Python-based web project "In Space", 2021), QWER Hacks (medication tracking Android application, 2021-2022)
- **Software:** Jira, Jama, AutoCAD, Git, Perforce, Kubernetes, Figma, Blender, Unity, Arduino Electronics, Adobe Suite, Office 365, Benchling, Bootstrap, Wordpress
- **Spoken Languages:** English, French, Polish (intermediate)

LEADERSHIP:

Electrical & Computer Engineering 88S: The Internet and Free Speech Seminar

Course Instructor | September 2021 - June 2022

- Devised and taught a 10-week course for UCLA undergraduate students, delving into the broader implications of computer engineering on democracy and global free speech; achieved average student rating of 4.96/5.0

UCLA Student Wellness Commission

Program Director and Curriculum Development Director | May 2020 - June 2022

Cirque de 'LA

Founder and President | September 2018 - June 2023

- Founded first organization at UCLA providing low-cost acrobatics instruction to over 300 community members; led board members in meetings, directed Spring Sing performance for an audience of 7000+